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Hospital Management System using Web Technology

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Abstract

Today's web-based technology offers many online services in almost every field. Every major industry is converting and establishing a digital front for all their major operations to get closer to the booming digital market. In today's world the information flow is very fast and redundant means will not add to the betterment of the individual or the organisation that is using them. Online connectivity is now a must for all well organised and well managed establishments. One such field is healthcare where the digitisation of information should happen rapidly and efficiently. This paper focuses on a web-based platform that replaces paper prescriptions with electronic medical management, improving patient management, doctor scheduling, and access to patient data within the hospital. [file:1]

Keywords: Hospital Management; Database; Webpage; Interface

1. Introduction

Many industries are turning towards the digital front, and it is beneficial for hospitals to move in the same direction. The existing manual module is efficient only when time is not a constraint, but in healthcare each second matters. The system should include online features such as patient records, disease history, and reports that doctors can access from anywhere through a database server. [file:1]

When a patient is admitted, vital details are updated for doctors to check online, and prescriptions can be sent directly to the pharmacy using a patient ID. Website visitors can register themselves as patients, obtain a unique patient ID, book appointments, check doctor availability, receive notifications for regular check-ups and medicine reorders, and access tips for a better lifestyle and health. [file:1]

Nowadays almost every individual has a smart device connected to the internet, so the speed and availability of data become critical. This digital approach helps people who need medical services for minor inconveniences and are unable to travel for necessary treatment. The paper is organised as follows: Section 2 presents the literature survey, Section 3 the objectives, Section 4 the proposed system, Section 5 the methodology, Section 6 the results and discussion, Section 7 the conclusions, and Section 8 the future enhancements. [file:1]

2. Literature Survey

The design of the project is influenced by the paper "Functional Description of Online Medical Management System Using Modern Technology" by Priyanka Patil, Sruthi Kunhiraman, and Rohini Temkin, which helped define the core online medical management functions. The paper "Intelligent Hospital Management System (IHMS)" by B. Koyuncu and H. Koyuncu informed the kinds of tasks to be handled by the system. The work of Z. Liu on hospital emergency nursing information management provided ideas about overall system design. [file:1]

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changing trends in the field. [file:1]

3. Objectives

The primary objective is to define, implement, and build a system that offers support for hospital management. This is achieved through several secondary objectives, such as improving efficiency of use and ensuring a consistent graphical user interface. A user is considered efficient when the time required to complete a task decreases with each usage. [file:1]

Another objective is to create a system that supports future improvements and functional extensions. The system should provide capabilities to manage patient and doctor details, schedule appointments, view prescriptions, order medicines online, and make online payments. [file:1]

4. Proposed System

The existing system is not completely digitised; most processes such as patient registration, sharing reports, and issuing prescriptions are offline, require significant paperwork, and consume time. The proposed project focuses on reducing paperwork and time, and integrates a predictor module that can predict the disease of a patient. [file:1]

The project is designed with two main modules: patient and doctor. The patient module is a web interface that allows users to register on the hospital website, schedule appointments, make payments, and view reports online. Doctor appointments are automated using the ID3 algorithm so that when a patient requests a doctor, an available doctor is automatically allotted. A disease prediction section analyses symptoms entered by the patient. [file:1]

The doctor module is another web interface where a doctor can view assigned patients, examine their history, and add comments about each visit. There are submodules for pharmacy and lab staff to view payments, share reports, and manage bills. An admin module allows viewing of all patient and doctor details and the addition and updating of staff records. [file:1]

5. Methodology

Two major modules are identified: patient and doctor. The patient module is used to book appointments, make payments, view reports, and review medical and payment history. It also contains a disease prediction section for patients. The doctor module provides the user interface for employees to access the database. [file:1]

In the doctor module there are four types of users: Admins, Doctors, Lab staff, and Pharmacy staff, each with hierarchical access to the database. Admins add users and assign access based on their designation. Doctors can view patient details, prescribe medicines, and see assigned tests. Lab staff manage the payment section and reports. Pharmacy staff add or remove medicine details and dispatch medicines based on payment information. [file:1]

Figure 1: Flow diagram describing the doctor module (confirming appointments and sharing reports) and the patient module (registration, secure login, booking appointments, and viewing reports). [file:1]

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6. Results and Discussion

Figure 3: Pharmacy staff dashboard – main page shows the staff profile. The patient tab contains an "add price" page to confirm prices and share bills with patients, and a dispatch page to confirm medicine dispatch. The medicine tab allows updates to prices and stock of existing medicines and the addition of new medicines. [file:1]

Figure 4: Admin dashboard – main page displays the admin profile. An "add" tab allows the admin to add new doctors and other staff and approve users. A "view" tab allows viewing and modification of patient, doctor, and staff details. [file:1]

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Figure 4: Admin dashboard – main page displays the admin profile. An "add" tab allows the admin to add new doctors and other staff and approve users. A "view" tab allows viewing and modification of patient, doctor, and staff details. [file:1]

Figure 5: Doctor dashboard – main page displays the doctor profile. The patients page allows the doctor to confirm appointments, share reports with patients, and view patient history. [file:1]

Figure 6: Patient dashboard – main page shows the patient profile. A "My orders" tab lists all medical or lab orders related to the patient. The patient tab allows viewing other profiles and booking appointments. The payment page enables additional payments for orders and adding money to an online wallet. [file:1]

7. Conclusions

A hospital management system is essential for maintaining details about doctors, patients, and hospital staff. Introducing such a system makes hospital work more seamless and efficient. Transferring patient data takes only seconds compared to manually sending physical files, and routine accounting tasks can be automated and simplified. [file:1]

8. Future Additions

The current system uses basic algorithms like ID3 for doctor allotment and disease prediction. Future work can integrate more complex algorithms to improve overall system performance and predictive accuracy. [file:1]

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